

A latent clinical-anatomical dimension relating metabolic syndrome to brain structure and cognition

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Abstract

The link between metabolic syndrome (MetS) and neurodegenerative as well cerebrovascular conditions holds substantial implications for brain health in at-risk populations. This study elucidates the complex relationship between metabolic syndrome (MetS) and brain health by conducting a comprehensive examination of cardiometabolic risk factors, cortical morphology, and cognitive function in 40,087 individuals. Multivariate, data-driven statistics identified a latent dimension linking more severe MetS to widespread cortical abnormalities and lower cognitive performance, accounting for up to 77% of shared variance in the data. This dimension was replicable across sub-samples. Our results also suggest that MetS-related cortical effects are shaped by the regional cellular composition and macroscopic brain network organization. By leveraging extensive, multi-domain data combined with a dimensional stratification approach, our analysis provides profound insights into the association of MetS and brain health. These findings underscore the necessity for effective risk mitigation strategies aimed at maintaining brain integrity.

eLife assessment

This **important** study enhances our understanding of the relationship between metabolic syndrome (MetS) and brain health from two large-scale datasets and crosses different scales of investigation. The results range from **solid** to **inadequate**, with the overall effects of MetS on the brain well supported, but the claimed inference of non-fasting blood glucose reflecting insulin resistance and suggestions of causative link to cognitive function need to be revised or tempered. Overall this study will be of great interest to researchers and clinicians seeking to understand metabolic syndrome.

1 Introduction

Metabolic syndrome (MetS) represents a cluster of cardiometabolic risk factors, including abdominal obesity, arterial hypertension, dyslipidemia, and insulin resistance [1]. With a prevalence of 23-35% in Western societies, it poses a considerable health challenge, promoting neurodegenerative and cerebrovascular diseases such as cognitive decline, dementia, and stroke [2–6]. As lifestyle and pharmacological interventions can modify the trajectory of MetS, advancing our understanding of its pathophysiological effects on brain structure and function as potential mediators of MetS-related neurological diseases is crucial to inform and motivate risk reduction strategies [7].

Magnetic resonance imaging (MRI) is a powerful non-invasive tool for examining the intricacies of neurological conditions in vivo. Among studies exploring MetS and brain structure, one of the most consistent findings has been alterations in cortical grey matter morphology [8]. Still, our understanding of the relationship between MetS and brain structure is constrained by several factors. To date, there have been only few studies on MetS effects on cortical grey matter integrity that are well-powered [9–12]. The majority of analyses are based on small sample sizes and report effects only on global measures of brain morphology or a priori-defined regions of interest, limiting their scope [12–14]. As a result, reported effects are heterogeneous and most likely difficult to reproduce [15]. Existing large-scale analyses on the isolated effects of individual risk factors (such as hypertension or obesity) do not account for the high covariance of MetS components driven by interacting pathophysiological effects, which may prevent them from capturing the whole picture of MetS as a risk factor composite [16–19]. In addition, analyses addressing the complex interrelationship of MetS, brain structure and cognitive functioning by investigating them in conjunction are scarce [8]. Lastly, while previous studies adopted a case-control design treating MetS as a broad diagnostic category [10–12], a dimensional approach viewing MetS as a continuum could offer a more nuanced representation of the multivariate, continuous nature of the risk factor composite.

Despite reports on MetS effects on cortical structure, the determinants and spatial effect patterns remain unclear. A growing body of evidence shows that spatial patterns of brain pathology are shaped by multi-scale neurobiological processes, ranging from the cellular level to regional dynamics to large-scale brain networks [20]. Accordingly, disease effects can not only be driven by local properties, when local patterns of tissue composition predispose individual regions to pathology, but also by topological properties of structural and functional brain networks [20, 21]. Guided by these concepts, multi-modal analysis approaches could advance our understanding of the mechanisms influencing MetS effects on cortical morphology.

We argue that further research leveraging extensive clinical and brain imaging data is required to explore MetS effects on cortical morphology. These examinations should integrate 1) a research methodology that strikes a balance between resolving the multivariate connection of MetS and brain structure while accounting for the high covariance of MetS components; 2) the recognition of impaired cognitive function as a pertinent consequence of MetS; and 3) the analysis of the spatial effect pattern of MetS and its possible determinants.

To meet these research needs, we investigated cortical thickness measurements in a pooled sample of two large-scale population-based cohorts from the UK Biobank (UKB) and Hamburg City Health Study (HCHS) comprising in total 40,087 participants. Partial least squares correlation analysis (PLS) was employed to characterize MetS effects on regional cortical morphology. PLS is especially suitable for this research task as it identifies overarching latent relationships by establishing a data-driven multivariate mapping between MetS components and cortical thickness. Furthermore, capitalizing on the cognitive phenotyping of both investigated cohorts, we examined the interrelation between MetS, cognitive function and cortical thickness. Finally, to uncover the determinants underlying differential MetS effects on cortical areas, we mapped local cellular as well as network topological attributes to observed MetS-associated cortical abnormalities. With this work, we aimed to advance the understanding of the fundamental principles underlying the neurobiology of MetS.

2 Materials and methods

2.1 Study population – the UK Biobank and Hamburg City Health Study

Here, we investigated cross-sectional clinical and imaging data from two large-scale population-based cohort studies: 1) the UK Biobank (UKB, $n = 39,668$, age 45–80 years; application number 41655) and 2) the Hamburg City Health Study (HCHS, $n = 2637$, age 45–74 years) [22,23]. Both studies recruit large study samples with neuroimaging data alongside detailed demographic and clinical assessment. Respectively, data for the first visit including a neuroimaging assessment were included. Individuals were excluded if they had a history or a current diagnosis of neurological or psychiatric disease. UKB individuals were excluded based on the non-cancer illnesses codes (<http://biobank.ndph.ox.ac.uk/showcase/coding.cgi?id=6>). Excluded conditions were Alzheimer's disease; alcohol, opioid and other dependencies; amyotrophic lateral sclerosis; brain injury; brain abscess; chronic neurological problem; encephalitis; epilepsy; haemorrhage; head injury; meningitis; multiple sclerosis; Parkinson's disease; skull fracture. Same criteria were applied on HCHS individuals based on the neuroradiological evaluation and self-reported diagnoses variables. To enhance comparability to previous studies we supplemented a case-control analysis enabling to complement continuous multivariate statistical analyses by group statistics. Therefore, a MetS sample was identified based on the consensus definition of the International Diabetes Federation (supplementary text S1) and matched to a control cohort.

2.2 Ethics approval

The UKB was ethically approved by the North West Multi-Centre Research Ethics Committee (MREC). Details on the UKB Ethics and Governance framework are provided online (<https://www.ukbiobank.ac.uk/media/0xsbmfmw/egf.pdf>). The HCHS was approved by the local ethics committee of the Landesärztekammer Hamburg (State of Hamburg Chamber of Medical Practitioners, PV5131). Good Clinical Practice (GCP), Good Epidemiological Practice (GEP) and the Declaration of Helsinki were the ethical guidelines that governed the conduct of the HCHS [24]. Written informed consent was obtained from all participants investigated in this work.

2.3 Clinical assessment

In the UK Biobank, a battery of cognitive tests is administered, most of which represent shortened and computerized versions of established tests aiming for comprehensive and concise assessment of cognition [25]. From this battery we investigated tests for executive function and processing speed (Reaction Time Test, Symbol Digit Substitution Test, Tower Rearranging Test, Trail Making Tests parts A and B), memory (Numeric Memory Test, Paired Associate Learning Test, Prospective Memory Test) and reasoning (Fluid Intelligence Test, Matrix Pattern Completion Test). Detailed descriptions of the individual tests can be found elsewhere [26]. Furthermore, some tests (Matrix Pattern Completion Test, Numeric Memory Test, Paired Associate Learning Test, Symbol Digit Substitution Test, Trail Making Test and Tower Rearranging Test) are only administered to a subsample of the UKB imaging cohort explaining the missing test results for a subgroup of participants.

In the HCHS, cognitive testing was administered by a trained study nurse and included the Animal Naming Test, Trail Making Test Part A and B, Verbal Fluency and Word List Recall subtests of the Consortium to Establish a Registry for Alzheimer's Disease Neuropsychological Assessment Battery (CERAD-Plus), as well as the Clock Drawing Test [27,28].

2.4 MRI acquisition

The full UKB neuroimaging protocol can be found online (https://biobank.ctsu.ox.ac.uk/crystal/crystal/docs/brain_mri.pdf) [22]. MR images were acquired on a 3-T Siemens Skyra MRI scanner (Siemens, Erlangen, Germany). T1-weighted MRI used a 3D MPRAGE sequence with 1-mm isotropic resolution with the following sequence parameters: repetition time = 2000 ms, echo time = 2.01 ms, 256 axial slices, slice thickness = 1 mm, and in-plane resolution = 1 x 1 mm. In the HCHS, MR images were acquired as well on a 3-T Siemens Skyra MRI scanner. Measurements were performed with a protocol as described in previous work [24]. In detail, for 3D T1-weighted anatomical images, rapid acquisition gradient-echo sequence (MPRAGE) was used with the following sequence parameters: repetition time = 2500 ms, echo time = 2.12 ms, 256 axial slices, slice thickness = 0.94 mm, and in-plane resolution = 0.83 x 0.83 mm.

2.5 Estimation cortical thickness

To achieve comparability and reproducibility, the preconfigured and containerized CAT12 pipeline (CAT12.7 r1743; <https://github.com/m-wierzbica/cat-container>) was harnessed for surface reconstruction and cortical thickness measurement building upon a projection-based thickness estimation method [29]. Cortical thickness measures were normalized from individual to 32k fsLR surface space (conte69) to ensure vertex correspondence across subjects. Individuals with a CAT12 image quality rating lower than 75% were excluded during quality assessment. To facilitate large-scale data management while ensuring provenance tracking and reproducibility, we employed the DataLad-based FAIRly big workflow for image data processing [30].

2.6 Statistical analysis

Statistical computations and plotting were performed in python 3.9.7 leveraging bctpy (v. 0.6.0), brainstat (v. 0.3.6), brainSMASH (v. 0.11.0), the ENIGMA toolbox (v. 1.1.3), matplotlib (v. 3.5.1), neuromaps (v. 0.0.1), numpy (v. 1.22.3), pandas (v. 1.4.2), pingouin (v. 0.5.1), pyls (v. 0.0.1), scikit-learn (v. 1.0.2), scipy (v. 1.7.3), seaborn (v. 0.11.2) as well as in matlab (v. 2021b) using ABAnnotate (v. 0.1.1).

2.6.1 Partial least squares correlation analysis

To relate MetS components and cortical morphology, we performed a PLS using pyls (<https://github.com/rmarkello/pyls>). PLS identifies covariance profiles that relate two sets of variables in a data-driven double multivariate analysis [31]. Here, we related regional cortical thickness measures

to clinical measurements of MetS components, i.e., obesity (waist circumference, hip circumference, waist-hip ratio, body mass index), arterial hypertension (systolic blood pressure, diastolic blood pressure), dyslipidemia (high density lipoprotein, low density lipoprotein, total cholesterol, triglycerides) and insuline resistance (blood glucose). Before conducting the PLS, missing values were imputed via k-nearest neighbor imputation (nneighbor = 4) with imputation only taking into account variables of the same group, i.e., MetS component variables were imputed based on the remaining MetS component data only and not based on demographic variables. To account for age, gender, education and cohort (UKB/HCHS) as potential confounds, they were regressed out of cortical thickness and MetS component data.

We then performed PLS as described in previous work [32]. Methodological details are covered in **figure 1a** and supplementary text S2. Cortical thickness measures were randomly permuted (nrandom = 5000) to assess statistical significance of derived latent variables and their corresponding covariance profiles. Subject-specific PLS scores were computed, where higher scores signify stronger adherence of an individual to an identified covariance profile. Bootstrap resampling (n = 5000) was performed to assess the contribution of individual variables to the thickness-clinical relationship. Confidence intervals (95%) of singular vector weights were computed for clinical variables to assess the significance of their contribution. To estimate contributions of cortical regions, bootstrap ratios were computed as the singular vector weight divided by the bootstrap-estimated standard error. A high bootstrap ratio is indicative of a region's contribution, as a relevant region shows a high singular vector weight alongside a small standard error implying stability across bootstraps. The bootstrap ratio equals a z-score in case of a normally distributed bootstrap. Hence, brain region contributions were considered significant if the bootstrap ratio was >1.96 or <-1.96 (95% confidence interval). Overall model robustness was assessed via a 10-fold cross-validation by correlating out-of-sample PLS scores within each fold.

2.6.2 Associations of cognitive and imaging data

As cognitive assessment differed between UKB and HCHS, cognitive variables were not included in the main PLS analysis. To further explore the interrelationship of cortical thickness and cognitive measures, we reconducted the PLS on the individual cohorts including cognitive measures to the clinical variables.

2.6.3 Contextualization analysis

We investigated the impact of MetS on cortical thickness in the context of cell-specific gene expression profiles and structural and functional brain network characteristics. (**figure 1b-d**). Therefore, we Schaefer-parcellated (400×7 and 100×7, v.1) the bootstrap ratio map and related it to indices representing different gene expression and network topological properties of the human cortex via spatial correlations (Spearman correlation, r_{sp}) on a group-level [33].

Virtual histology analysis. We performed a virtual histology analysis leveraging gene transcription information to quantify the density of different cell populations across the cortex. Genes corresponding with specific cell populations of the central nervous system were identified based on a classification derived from single nucleus-RNA sequencing data [34]. The gene-celltype mapping is provided by the PsychENCODE database (http://resource.psychencode.org/Datasets/Derived/SC_Decomp/DER-19_Single_cell_markergenes_TPM.xlsx) [35]. The abagen toolbox (v. 0.1.3) and the ENIGMA toolbox (v. 1.1.3) were used to obtain regional microarray expression data of these genes for Schaefer100×7 parcels based on the Allen Human Brain Atlas (AHBA) [36]. The Schaefer100×7 atlas was used as it better matches the sampling density of the AHBA eventually resulting in no parcels with missing values. Regional expression patterns of genes corresponding to astrocytes, endothelial cells, excitatory neuron populations (Ex1-8), inhibitory neuron populations (In1-8), microglia, and oligodendrocytes were extracted. Instead of assessing the correspondence between MetS effects and the expression pattern of each gene directly, we employed ensemble-based gene category enrichment analysis (GCEA) as implemented in the

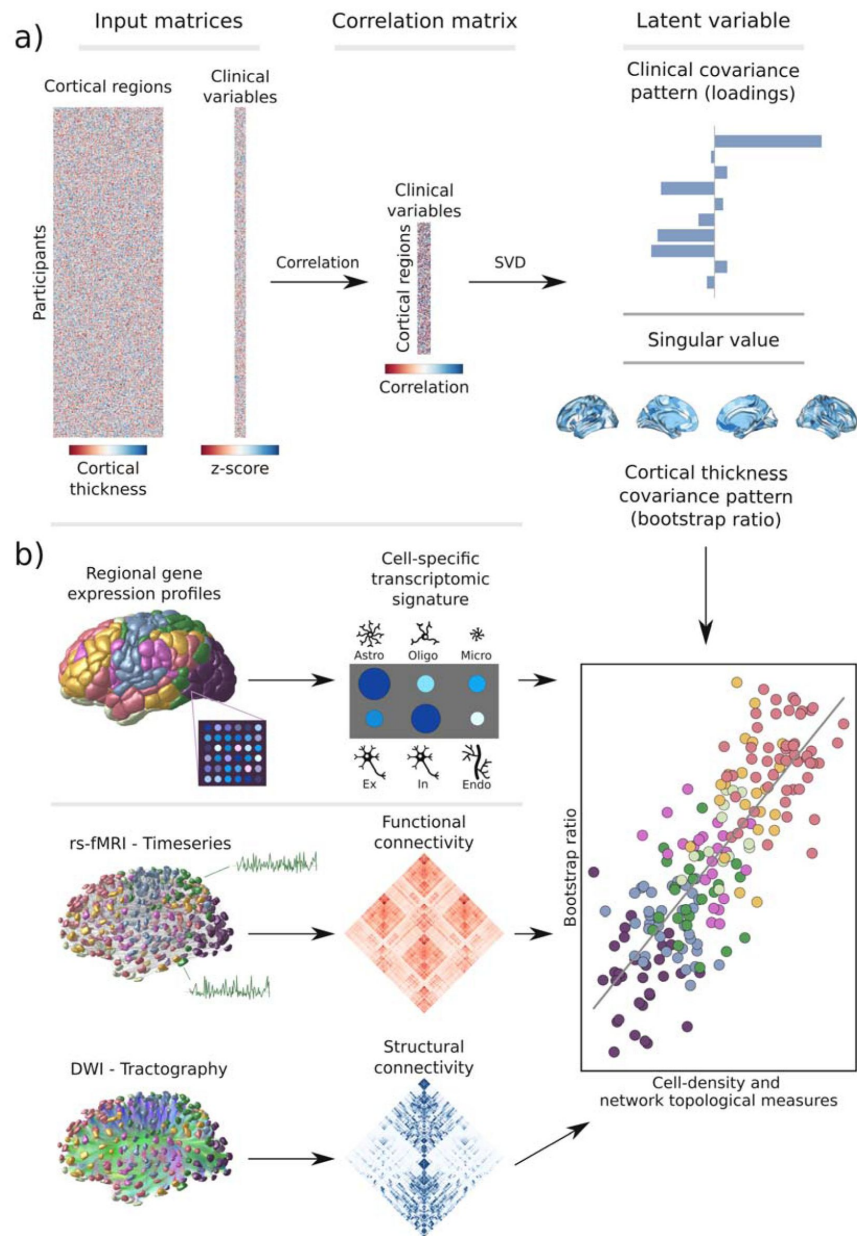


Figure 1.

Methodology.

a) Illustration of the partial least squares correlation analysis. Starting from two input matrices containing per-subject information of regional cortical thickness measures as well as clinical data (demographic and MetS-related risk factors) a correlation matrix is computed. This is subsequently subjected to singular value decomposition resulting in a set of mutually orthogonal latent variables. Latent variables each consist of a left singular vector (here, clinical covariance pattern), singular value and right singular vector (here, cortical thickness covariance profile). b-d) Contextualization analysis. b) Based on microarray gene expression data, the densities of different cell populations across the cortex were quantified. c) Based on functional and structural group-consensus connectomes based on data from the Human Connectome Project, metrics of functional and structural brain network topology were derived. d) Cell density as well as connectomic measures were related to the bootstrap ratio via spatial correlations. Modified from Petersen et al. and Zeighami et al. [32,76]. Abbreviations: Astro – astrocytes; DWI – diffusion-weighted magnetic resonance imaging; Endo – endothelial cells; Ex – excitatory neuron populations (Ex1-8); In – inhibitory neuron populations (In1-8); Micro – microglia; Oligo – oligodendrocytes; rs-fMRI – resting-state functional magnetic resonance imaging; SVD – singular value decomposition.

ABAnnotate toolbox [37,38]. This approach represents a modification to customary GCEA addressing the issues of gene-gene dependency through within-category co-expression which is caused by shared spatial embedding as well as spatial autocorrelation of cortical transcriptomics data [39]. In brief, gene transcription indices were averaged within categories (here cell populations) and spatially correlated with the bootstrap ratio map. Statistical significance was assessed by comparing the empirical correlation coefficients against a null distribution derived from surrogate maps with preserved spatial embedding and autocorrelation computed via a spatial lag model [40].

Brain network topology. To investigate the cortical MetS effects pattern in the context of brain network topology, three connectivity metrics were leveraged based on data from structural and functional brain imaging: weighted degree centrality, neighborhood abnormality as well as macroscale functional connectivity gradients as described previously [32]. These were computed based on functional and structural consensus connectomes on group-level derived from the Human Connectome Project Young Adults dataset comprised in the ENIGMA toolbox [41,42]. Computation and derivation of the metrics are described in the supplementary text S3. For this analysis, statistical significance of spatial correlations was assessed via spin permutations ($n = 1,000$) which represent a null model preserving the inherent spatial autocorrelation of cortical information [43]. Spin permutations are performed by projecting parcel-wise data onto a sphere which then is randomly rotated. After rotation, information is projected back on the surface and a permuted r_{sp} is computed. A p-value is computed comparing the empirical correlation coefficient to the permuted distribution. To assure that our results do not depend on null model choice, we additionally tested our results against a variogram-based null model implemented in the brainSMASH toolbox (<https://github.com/murraylab/brainsmash>) as well as a network rewiring null model with preserved density and degree sequence [44,45].

All p-values resulting from both contextualization analyses were FDR-corrected for multiple comparisons. As we conducted this study mindful of the reuse of our resources, the MetS effect maps are provided as separate supplementary files to enable further analyses.

2.6.4 Sensitivity analyses

To test whether the PLS indeed captures the effect of MetS on cortical thickness, we conducted a group comparison as in previous studies of MetS. Besides descriptive group statistics, individuals with MetS and matched controls were compared on a surface vertex-level leveraging the BrainStat toolbox (v 0.3.6, <https://brainstat.readthedocs.io/>) [46]. A general linear model was applied correcting for age, gender, education and cohort effects. Vertex-wise p-values were FDR-corrected for multiple comparisons. To demonstrate the correspondence between the t-statistic and bootstrap ratio maps, we related them via spatial correlation analyses. The t-statistic map was also used to sensitivity analyze the virtual histology analysis and brain network contextualization.

To ensure that the brain network contextualization results were not biased by the connectome choice, we reperformed the analysis with structural and functional group consensus connectomes based on resting-state functional and diffusion-weighted MRI data from the Hamburg City Health Study. The corresponding connectome reconstruction approaches were described elsewhere [32].

3 Results

Sample characteristics

Application of exclusion criteria and quality assessment ruled out 2,188 UKB subjects and 30 HCHS subjects resulting in a final analysis sample of 40,087 individuals. For a flowchart providing details on the sample selection procedure please refer to supplementary figure S4. Descriptive statistics are listed in **Table 1**. To sensitivity analyze our results, as well as to facilitate the comparison with previous reports which primarily rely on a case-control design, we supplemented group statistics comparing individuals with clinically defined MetS and matched controls, where applicable. Corresponding group analysis results are described in more detail in supplementary materials S5-11.

3.1 Partial least squares correlation

We investigated the relationship between cortical thickness and clinical measures of MetS (abdominal obesity, arterial hypertension, dyslipidemia, insulin resistance) in a PLS considering all individuals from both studies ($n=40,087$). By this, we aimed to detect the continuous effect of any MetS component independent from a formal binary classification of MetS (present / not present). A correlation matrix relating all considered MetS component measures is displayed in supplementary figure S12. Before conducting the PLS, cortical thickness and clinical data were deconfounded for age, gender, education and cohort effects. PLS identified seven significant latent variables which represent clinical-anatomical dimensions relating MetS components to cortical thickness (supplementary table S13). The first latent variable explained 77.48% of shared variance and was thus further investigated (**figure 2a**). Specifically, the first latent variable corresponded with a covariance pattern of lower severity of MetS (**figure 2b**; loadings [95% confidence interval]; waist circumference: $-.228 [-.237, -.219]$, hip circumference: $-.188 [-.197, -.179]$, waist-hip ratio: $-.162 [-.171, -.152]$, body mass index: $-.230 [-.239, -.221]$, systolic blood pressure: $-.098 [-.108, -.089]$, diastolic blood pressure: $-.121 [-.131, -.112]$, high density lipoprotein: $.096 [.087, .106]$, low density lipoprotein: $-.013 [-.023, -.004]$, total cholesterol: $.002 [-.008, .011]$, triglycerides: $-.092 [-.101, -.082]$, glucose: $-.049 [-.058, -.040]$). Notably, the obesity-related measures showed the strongest contribution to the covariance profile as indicated by the highest loading to the latent variable. Age ($.004 [-.006, .013]$), gender ($.004 [-.005, .014]$), education ($.002 [-.007, .010]$) and cohort ($.001 [-.008, .010]$) did not significantly contribute to the latent variable, which is compatible with sufficient effects of deconfounding. Details on the second latent variable which explained 17.26% of shared variance are provided in supplementary figure S14.

Bootstrap ratios ($= \frac{\text{singular vector weight}}{\text{bootstrap-estimated standard error}}$) were computed to identify cortical regions with a significant contribution to the covariance profile (see Methods). Cortical thickness in orbitofrontal, lateral prefrontal, insular, anterior cingulate and temporal areas contributed positively to the covariance pattern as indicated by a positive bootstrap ratio (**figure 2c**). Thus, a higher cortical thickness in these areas corresponded with less obesity, hypertension, dyslipidemia and insulin resistance and vice versa, i.e., lower cortical thickness with increased severity of MetS. A negative bootstrap ratio was found in superior frontal, parietal and occipital regions indicating that a higher cortical thickness in these regions corresponded with a more pronounced expression of MetS components. This overall pattern was confirmed via conventional, vertex-wise group comparisons of cortical thickness measurements based on the binary classification of individuals with MetS and matched controls (supplementary figure S11) as well as subsample analyses considering the UKB and HCHS participants independently (supplementary figure S15-16). The correlation matrix of all spatial effect maps investigated in this study (bootstrap ratio and Schaefer400-parcellated t-statistic from group comparisons) is visualized in supplementary figure S17. All derived effect size maps were significantly correlated ($r_{sp} = .70 - .99$, $p_{FDR} < .05$) [33].

Metric	Stat ^a
Age (years)	63.55 ± 7.59 (40087)
Gender (% female)	46.47 (40087)
Education (ISCED)	2.62 ± 0.73 (39944)
<u>Metabolic syndrome components</u>	
Waist circumference (cm)	88.47 ± 12.71 (38800)
Hip circumference (cm)	100.90 ± 8.79 (38801)
Waist-hip ratio	0.88 ± 0.09 (38800)
Body mass index	26.47 ± 4.37 (38701)
RR _{systolic} (mmHg)	138.30 ± 18.57 (31234)
RR _{diastolic} (mmHg)	78.88 ± 10.09 (31238)
Antihypertensive therapy (%)	6.96 (39976)
HDL (mg/dL)	61.76 ± 23.69 (34468)
LDL (mg/dL)	137.38 ± 36.29 (37456)
Cholesterol (mg/dL)	211.29 ± 56.42 (37531)
Triglycerides (mg/dL)	148.90 ± 83.84 (37510)
Lipid lowering therapy (%)	14.44 (39976)
Blood glucose (mg/dL)	90.29 ± 17.58 (34432)
Antidiabetic therapy (%)	0.45 (39976)

Table 1.

Descriptive statistics UKB and HCHS

<u>Imaging</u>	
Mean cortical thickness (mm)	2.40 ± 0.09 (40087)
<u>Cognitive variables of the UK Biobank</u>	
Fluid Intelligence	6.63 ± 2.06 (36510)
Matrix Pattern Completion	7.99 ± 2.13 (25771)
Numeric Memory Test	6.69 ± 1.52 (26780)
Paired Associate Learning	6.92 ± 2.63 (26048)
Prospective Memory	1.07 ± 0.39 (37192)
Reaction Time (sec)	594.16 ± 109.08 (37015)
Symbol Digit Substitution	18.96 ± 5.25 (25810)
Tower Rearranging Test	9.91 ± 3.23 (25555)
Trail Making Test A (sec)	223.03 ± 86.51 (26048)
Trail Making Test B (sec)	550.01 ± 270.09 (26048)
<u>Cognitive variables of the Hamburg City Health Study</u>	
Animal Naming Test	24.78 ± 6.92 (2416)
Clock Drawing Test	6.43 ± 1.12 (2479)
Trail Making Test A (sec)	40.09 ± 14.33 (2290)
Trail Making Test B (sec)	90.05 ± 37.30 (2264)
Multiple-Choice Vocabulary Intelligence Test	31.27 ± 3.58 (2026)
Word List Recall	7.75 ± 1.84 (2342)
Abbreviations: cm = centimeter, dL = deciliter, HDL = high density lipoprotein, ISCED = International Standard Classification of Education, mg = milligram, mm = millimeters, mmHg = millimeters of mercury,	

Table 1. (continued)

RR = Blood pressure, sec = seconds

^aPresented as mean $\pm$ SD (N)

Table 1. (continued)

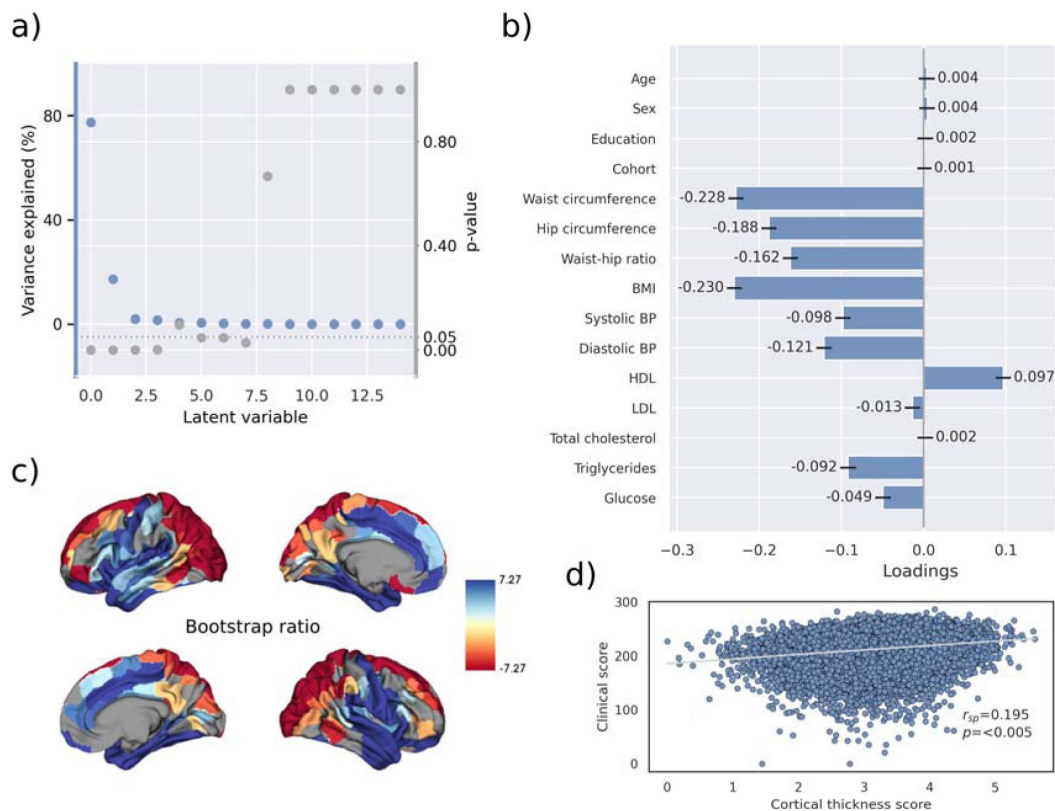


Figure 2.

Partial least squares (PLS) analysis.

a) Explained variance and p-values of latent variables. b) Clinical covariance profile. 95% confidence intervals were calculated via bootstrap resampling. Note that confound removal for age, gender, education and cohort was performed prior to PLS. c) Bootstrap ratio representing the covarying cortical thickness pattern. A high positive or negative bootstrap ratio indicates high contribution of a brain region to the overall covariance pattern. Vertices with a significant bootstrap ratio (> 1.96 or < -1.96) are highlighted by colors. d) Scatter plot relating subject-specific clinical and cortical thickness scores. Higher scores indicate higher adherence to the respective covariance profile. Abbreviations: - Spearman correlation coefficient.

Subject-specific cortical thickness and clinical scores for the first latent variable were computed. These scores indicate to which degree an individual expresses the corresponding covariance profiles. By definition, the scores are correlated ($r_{sp} = .195$, $p < .005$, [figure 2d](#)) indicating that individuals exhibiting the clinical covariance pattern (severity of MetS components) also express the cortical thickness pattern. This relationship was robust across a 10-fold cross-validation (avg. $r_{sp} = .21$, supplementary table S18).

To examine the interrelationship of MetS components, cortical thickness and cognition, we conducted the PLS in the UKB sample and the HCHS sample, separately, as the cognitive assessment varied between UKB and HCHS. Within the individual cohorts, cognitive variables were included in the PLS as clinical variables alongside MetS components. The resulting first latent variable associated a more severe risk profile with cortical thickness abnormalities, but also worse cognitive performance as shown by respective loadings. Significant cognitive loadings in the UKB analysis were identified for the Fluid Intelligence Test (loadings [95% confidence interval]; .063 [.053,.072]), Matrix Pattern Completion Test (.054 [.044,.064]), Numeric Memory Test (.032 [.022,.041]), Paired Associate Learning Test (.051 [.041,.060]), Reaction Time Test (-.040 [-.049,-.030]), Symbol Digit Substitution Test (.040 [.030,.050]), Tower Rearranging Test (.035 [.025,.045]) and Trail Making Test Part B (-.019 [-.029,-.009]) (supplementary figure S15b). A significant HCHS-specific cognitive loading was found for the Multiple-choice Vocabulary Intelligence Test (.070 [.031,.109]) (supplementary figure S16b). Group comparisons of cognitive measures within the UKB and HCHS subsamples are provided in supplementary tables S8-9.

3.2 Contextualization of MetS-associated cortical thickness abnormalities

We investigated whether the pattern of MetS effects on cortical structure is conditioned by the regional density of specific cell populations and global brain network topology in a surface-based contextualization analysis (see Methods).

Therefore, we first used a virtual histology approach to relate the bootstrap ratio from PLS to the differential expression of cell-type specific genes based on microarray data from the Allen Human Brain Atlas [\[47\]](#). The results are illustrated in [figure 3](#). The bootstrap ratio was significantly positively correlated with the density of endothelial cells ($Z_{r_{sp}} = .192$ $p_{FDR} = .013$), microglia ($Z_{r_{sp}} = .268$ $p_{FDR} = .013$), excitatory neurons type 8 ($Z_{r_{sp}} = .166$ $p_{FDR} = .013$), excitatory neurons type 6 ($Z_{r_{sp}} = .156$ $p_{FDR} = .002$) and inhibitory neurons type 1 ($Z_{r_{sp}} = .368$ $p_{FDR} = .036$) indicating that MetS effects on cortical thickness are strongest in regions of the highest density of these cell types. No significant associations were found with regard to the remaining excitatory neuron types (Ex1-Ex5, Ex7), inhibitory neurons (In2-In8), astrocytes and oligodendrocytes (supplementary table S19). As a sensitivity analysis, we contextualized the t-statistic map derived from group statistics. The results remained stable except for excitatory neurons type 6 ($Z_{r_{sp}} = .145$ $p_{FDR} = .123$) and inhibitory neurons type 1 ($Z_{r_{sp}} = .432$ $p_{FDR} = .108$), which no longer showed a significant association (supplementary materials S20-21).

Second, we associated the bootstrap ratio with three pre-selected measures of brain network topology derived from group consensus functional and structural connectomes of the Human Connectome Project (HCP) ([figure 4](#)): weighted degree centrality (marking brain network hubs), neighborhood abnormality and macroscale functional connectivity gradients.[\[32\]](#) The bootstrap ratio showed a medium positive correlation with the functional neighborhood abnormality ($r_{sp} = .452$, $p_{spin} = .002$, $p_{smash} < .001$, $p_{rewire} < .001$) and a strong positive correlation with the structural neighborhood abnormality ($r_{sp} = .770$, $p_{spin} < .001$, $p_{smash} < .001$, $p_{rewire} < .001$) indicating functional and structural interconnectedness of areas experiencing similar MetS effects. These results remained significant when the t-statistic map was contextualized instead of the bootstrap ratio as well as when neighborhood abnormality measures were derived from consensus connectomes of the HCHS instead of the HCP (supplementary figure S22-23). We found no

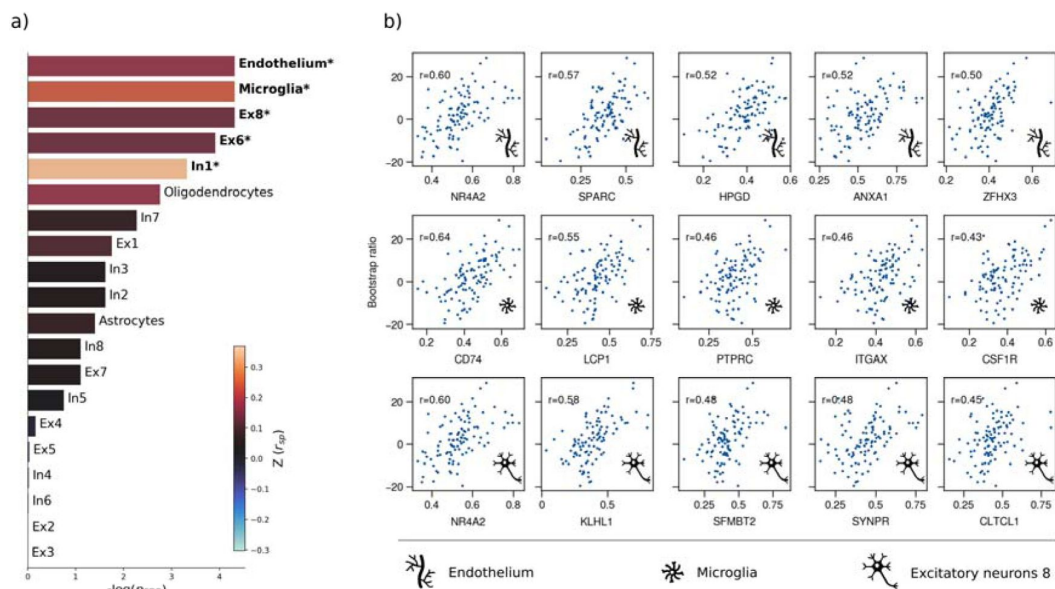


Figure 3.

Virtual histology analysis.

The correspondence between MetS effects (bootstrap ratio) and cell type-specific gene expression profiles was examined via an ensemble-based gene category enrichment analysis. a) Barplot displaying spatial correlation results. The bar height displays the significance level. Colors encode the aggregate z-transformed Spearman correlation coefficient relating the Schaefer100-parcellated bootstrap ratio and respective cell population densities. b) Scatter plots illustrating spatial correlations between MetS effects and exemplary cortical gene expression profiles per cell population significantly associated across analyses – i.e., endothelium, microglia and excitatory neurons type 8. Top 5 genes most strongly correlating with the bootstrap ratio map were visualized for each of these cell populations. Icons in the bottom right of each scatter plot indicate the corresponding cell type. A legend explaining the icons is provided at the bottom. First row: endothelium; second row: microglia; third row: excitatory neurons type 8. A corresponding plot illustrating the contextualization of the t-statistic derived from group statistics is shown in supplementary figure S20. Abbreviations: – negative logarithm of the false discovery rate-corrected p-value derived from spatial lag models [38, 40]; – Spearman correlation coefficient. – aggregate z-transformed Spearman correlation coefficient.

significant associations for the remaining indices of network topology, i.e., functional degree centrality ($r_{sp} = .162$, $p_{spin} = .362$, $p_{smash} = .362$, $p_{rewire} = .876$), structural degree centrality ($r_{sp} = .038$, $p_{spin} = .393$, $p_{smash} = .828$, $p_{rewire} = .082$) as well as functional cortical gradient 1 ($r_{sp} = .159$, $p_{spin} = .293$, $p_{smash} = .362$, $p_{rewire} = .018$) and gradient 2 ($r_{sp} = -.190$, $p_{spin} = .293$, $p_{smash} = .362$, $p_{rewire} = .001$).

4 Discussion

We investigated the impact of metabolic syndrome on cortical thickness and cognitive function in a large sample of individuals from two population-based neuroimaging studies. We report three main findings: 1) multivariate, data-driven statistics revealed a latent variable relating MetS and brain health: participants were distributed along a clinical-anatomical dimension of interindividual variability, linking more severe MetS to widespread cortical thickness abnormalities and lower cognitive performance; 2) negative MetS effects on cortical thickness were strongest in orbitofrontal, lateral prefrontal, insular, cingulate and temporal cortices; 3) this pattern of MetS-effects on cortical thickness appeared to be conditioned by regional cell composition as well as functional and structural connectivity. These findings were robust across sensitivity analyses. Collectively, our study provides a comprehensive assessment of the multifaceted relationship between MetS and cortical morphology.

MetS adversely impacts brain health through complex, interacting effects on the cerebral vasculature and parenchyma as shown by histopathological and imaging studies [19]. The pathophysiology of MetS involves atherosclerosis, which affects blood supply and triggers inflammation [48, 49]; endothelial dysfunction reducing cerebral vasoreactivity [50]; breakdown of the blood-brain barrier inciting an inflammatory response [51]; oxidative stress causing neuronal and mitochondrial dysfunction [52]; and small vessel injury leading to various pathologies including white matter damage, microinfarcts and cerebral microbleeds [53].

To address these interacting effects, we harnessed multivariate, data-driven statistics in form of a PLS in two large-scale population-based studies to probe for covariance profiles relating the full range of MetS components (such as obesity or arterial hypertension) to regional cortical thickness in a single analysis. PLS identified seven significant latent variables with the first variable explaining the majority (77.48%) of shared variance within the imaging and clinical data (**figure 2a**). This finding suggests that the overall relationship of MetS and brain structure is low-dimensional, i.e., effects of MetS components on cortical thickness are relatively uniform despite the diversity of underlying pathomechanisms. PLS revealed that all MetS components were contributing to this latent signature. However, waist circumference, hip circumference, waist-hip ratio and body mass index consistently contributed higher than the remaining variables across conducted analyses which highlights obesity as the strongest driver of MetS-related cortical morphometric abnormalities.

We interpret these findings as evidence that MetS-associated conditions jointly contribute to the harmful effects on brain structure rather than affecting it in a strictly individual manner. This notion is supported by previous work in the UKB demonstrating overlapping effects of individual risk factors on cortical morphology [54]. Specifically, the first latent variable related increased severity of obesity, dyslipidemia, arterial hypertension and insulin resistance with lower cortical thickness in orbitofrontal, lateral prefrontal, insular, cingulate and temporal cortices (**figure 2b** and 1c). This profile was consistent in separate PLS analyses of UKB and HCHS participants as well as group comparisons (supplementary figures S8 and S12-13). Previous research aligns with our detection of a MetS-associated frontotemporal morphometric abnormality pattern [9, 13, 55]. As a speculative causative pathway, human and animal studies have related the orbitofrontal, insular and anterior cingulate cortex to food-related reward processing, taste and

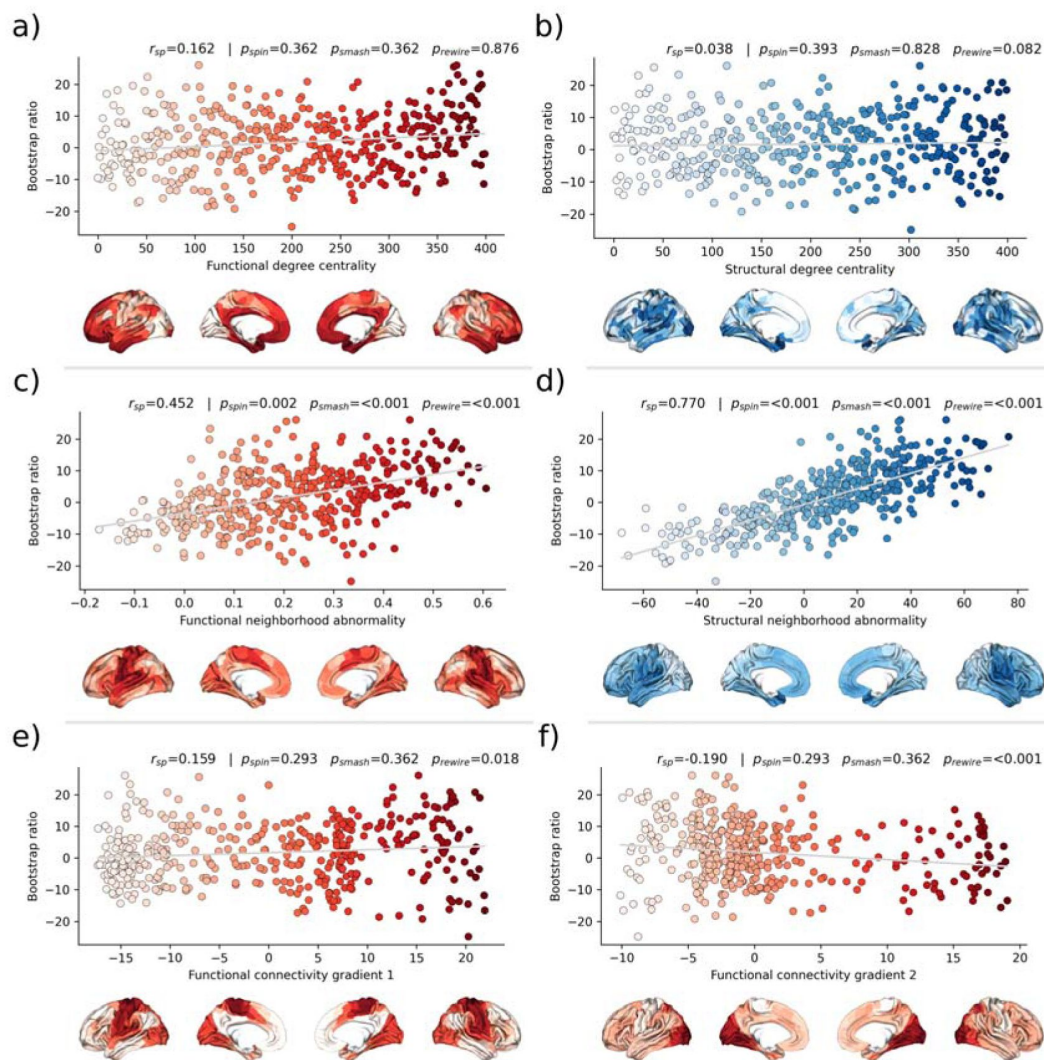


Figure 4.

Brain network contextualization.

Spatial correlation results derived from relating Schaefer400×7-parcellated cortical maps of MetS effects (bootstrap ratio) to network topological indices (red: functional connectivity, blue: structural connectivity). Scatter plots that illustrate the spatial relationship are supplemented by respective surface plots for anatomical localization. The color coding of cortical regions and associated dots corresponds. a) & b) Functional and structural degree centrality rank. c) & d) Functional and structural neighborhood abnormality. e) & f) Intrinsic functional network hierarchy represented by functional connectivity gradients 1 and 2. Complementary results concerning t-statistic maps derived from group comparisons between MetS subjects and controls are presented in supplementary figure S22. Abbreviations: p_{rewire} - p-value derived from network rewiring [45]; p_{smash} - p-value derived from brainSMASH surrogates [44]; p_{spin} - p-value derived from spin permutation results [43]; r_{sp} - Spearman correlation coefficient.

impulse regulation [55,56]. Conceivably, structural alterations of these brain regions are linked to brain functions and behaviors that exacerbate the risk profile leading to MetS [57,58]. Interestingly, we also observed a positive relationship between cortical thickness and MetS in the superior frontal, parietal and occipital lobe. Interpretation of this result is, however, less intuitive. We also noted a positive MetS-cortical thickness association in superior frontal, parietal and occipital lobes, a less intuitive finding that has been previously reported [60,61]. Although speculative, the positive effects might be due to MetS compensating cholesterol disruptions associated with neurodegenerative processes [62].

Cognitive performance has been consistently linked to cardiometabolic risk factors and structural brain abnormalities in health and disease [62]. Applying PLS to both datasets (UKB and HCHS) individually revealed that the identified latent variables were not only relating more severe MetS to cortical thickness abnormalities but also to worse cognitive performance (supplementary materials S15b and S16b). These associations were replicated by results from the group comparison highlighting worse performance in several cognitive domains in MetS (supplementary tables S8-9). As a potential mechanism underlying these associations, altered cortical structure has been shown to mediate the relationship between MetS and cognitive performance in a pediatric study and elderly patients with vascular cognitive impairment [64–66]. The detected latent variable might represent a continuous disease spectrum spanning from minor cognitive deficits due to a cardiometabolic risk profile to more severe vascular cognitive impairment and dementia. In support of this hypothesis, the determined cortical thickness abnormality pattern is consistent with the atrophy pattern found in vascular mild cognitive impairment and vascular dementia [65,66]. Collectively, we interpret these findings as evidence for an interrelationship between MetS, cognitive dysfunction and structural brain alteration.

To better understand the emergence of the spatial pattern of MetS effects on cortical thickness, we conducted two brain surface-based contextualization analyses leveraging reference datasets of (1) local gene expression data and (2) properties of brain network topology.

Using a virtual histology approach based on regional gene expression data, we investigated MetS effects in relation to cell population densities (**figure 3**). As the main finding, we report that higher MetS-related cortical abnormalities coincide with a higher regional density of endothelial cells. This aligns with the known role of endothelial dysfunction in MetS compromising tissues via chronic vascular inflammation, increased thrombosis risk and hypoperfusion due to altered vasoreactivity and vascular remodeling [50]. As endothelial density also indicates the degree of general tissue vascularization, well-vascularized regions are also likely more exposed to cardiometabolic risk factor effects in general [48]. Our results furthermore indicate that microglial density determines a brain region's susceptibility to MetS effects. Microglia are resident macrophages of the central nervous system sustaining neuronal integrity by maintaining a healthy microenvironment. Animal studies have linked microglia activation mediated by blood-brain barrier leakage and systemic inflammation to cardiometabolic risk [66,67]. Activated microglia can harm the brain structure by releasing reactive oxygen species, proinflammatory cytokines and proteinases [68]. Lastly, we found an association with the density of excitatory neurons of subtype 8. These neurons reside in cortical layer 6 and their axons mainly entertain long-range cortico-cortical and cortico-thalamic connections [34,70]. Consequently, layer 6 neurons might be particularly susceptible to MetS effects due to their exposition to MetS-related white matter disease [24,71]. Taken together, the virtual histology analysis indicates that MetS effects on cortical thickness are conditioned by local cellular fingerprints. Our results underscore the role of endothelial cells and microglia in cardiometabolic risk affecting the structural integrity of the cerebral cortex.

For the second approach, we contextualized MetS-effects on cortical thickness using principal topological properties of functional and structural brain networks. We found that regional MetS effects and those of functionally and structurally connected neighbors were correlated (**figure**

4d [\[70\]](#)) - i.e., areas with similar MetS effects tended to be disproportionately interconnected. Put differently, MetS effects coincided within functional and structural brain networks. Therefore, our findings provide evidence that a region's functional and structural network embedding – i.e., its individual profile of functional interactions as well as white matter fiber tract connections – determines its susceptibility to morphometric MetS effects. Multiple mechanisms might explain how connectivity conditions the cortical alterations related to MetS. For example, microvascular pathology might impair white matter fiber tracts leading to joint degeneration in interconnected cortical brain areas: that is, the occurrence of shared MetS effects within functionally and structurally connected neighborhoods is explained by their shared (dis-)connectivity profile [\[71\]](#). In support of this, previous work using diffusion tensor imaging suggests that MetS-related microstructural white matter alterations preferentially occur in the frontal and temporal lobe, which spatially matches the frontotemporal morphometric differences observed in our work [\[72\]](#). Furthermore, we speculate on an interplay between local and network-topological susceptibility in MetS: functional and structural connectivity may provide a scaffold for propagating MetS-related perturbation across the network in the sense of a spreading phenomenon - i.e., a region might be influenced by network-driven exposure to regions with higher local susceptibility. Observed degeneration of a region might be aggravated by malfunctional communication to other vulnerable regions including mechanisms of excitotoxicity, diminished excitation and metabolic stress [\[73\]](#).

While this work's strengths lie in a large sample size, high-quality MRI and clinical data, robust image processing, and a comprehensive methodology for examining MetS effects on brain health, it also has limitations. First, the virtual histology analysis relies on post-mortem brain samples, potentially different from in-vivo profiles. In addition, the predominance of UKB subjects may bias the results, and potential reliability issues of the cognitive assessment in the UK Biobank need to be acknowledged [\[74\]](#). Lastly, the cross-sectional design restricts the ability for demonstrating causative effects. Longitudinal assessment of the surveyed relationships would provide more robust evidence and therefore, future studies should move in this direction.

5 Conclusion

Our analysis revealed serious effects of MetS on brain health, complementing existing efforts to motivate and inform strategies for cardiometabolic risk reduction. In conjunction, a characteristic and reproducible structural imaging fingerprint associated with MetS was identified. This pattern of MetS-related cortical morphological abnormalities was governed by local histological as well as global network topological features. Collectively, our results highlight how an integrative, multi-modal analysis approach can lead to a more holistic understanding of the neural underpinnings of MetS and its risk components.

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Each author has made a significant contribution to the manuscript and all authors read and approved its final version. We describe individual contributions to the paper using the CRediT contributor role taxonomy. Conceptualization: M.P., F.H., B.C.; Data Curation: M.P., F.H., F.N., C.M., M.S.; Formal analysis: M.P.; Funding acquisition: S.B.E., G.T., B.C.; Investigation: M.P.; Methodology:

M.P.; Software: M.P., F.H.; Supervision: G.T., B.C.; Visualization: M.P.; Writing—original draft: M.P.; Writing—review & editing: M.P., F.H., F.N., C.M., M.S., L.R., B.Z., T.Z., S.K., J.G., J.F., R.T., A.O., K.P., S.B.E., G.T., B.C.

6 Declaration of interest

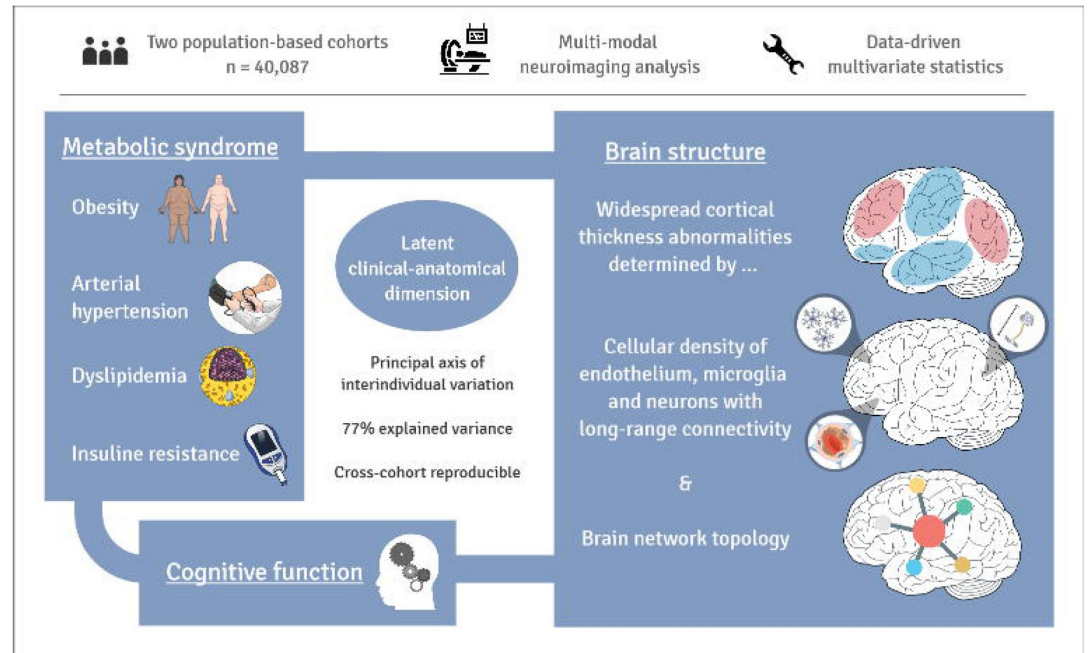
JG has received speaker fees from Lundbeck, Janssen-Cilag, Lilly, Otsuka and Boehringer outside the submitted work. JF reported receiving personal fees from Acandis, Cerenovus, Microvention, Medtronic, Phenox, and Penumbra; receiving grants from Stryker and Route 92; being managing director of eppdata; and owning shares in Tegus and Vastrax; all outside the submitted work. GT has received fees as consultant or lecturer from Acandis, Alexion, Amarin, Bayer, Boehringer Ingelheim, BristolMyersSquibb/Pfizer, Daichi Sankyo, Portola, and Stryker outside the submitted work. The remaining authors declare no conflicts of interest.

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8 Data availability

UK Biobank data can be obtained via its standardized data access procedure (<https://www.ukbiobank.ac.uk/>). HCHS participant data used in this analysis is not publicly available for privacy reasons but access can be established via request to the HCHS steering committee. The analysis code is publicly available on GitHub (https://github.com/csi-hamburg/2023_petersen_mets_cortical_thickness) and <https://github.com/csi-hamburg/CSIframe/wiki/Structural-processing-with-CAT>).



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Reviewer #1 (Public Review):

Summary:

In their study, Petersen et al. investigated the relationship between parameters of metabolic syndrome (MetS) and cortical thickness using partial least-squares correlation analysis (PLS) and performed subsequently a group comparison (sensitivity analysis). To do this, they utilized data from two large-scale population-based cohorts: the UK BioBank (UKB) and the Hamburg City Health Study (HCHS). They identified a latent variable that explained 77% of the shared variance, driven by several measures related to MetS, with obesity-related measures having the strongest contribution. Their results highlighted that higher cortical thickness in the orbitofrontal, lateral prefrontal, insular, anterior cingulate, and temporal areas is associated with lower MetS metric severity. Conversely, the opposite pattern was observed in the superior frontal, parietal, and occipital regions. A similar pattern was then observed in the sensitivity analysis when comparing two groups (MetS vs. matched controls) separately. They then mapped local cellular and network topological attributes to the

observed cortical changes associated with MetS. This was achieved using cell-type-specific gene expressions from the Allen Human Brain Atlas and the group consensus functional and structural connectomes of the Human Connectome Project (HCP), respectively. This contextualization analysis allowed them to identify potential cellular contributions in these structures driven by endothelial cells, microglial cells, and excitatory neurons. It also indicated functional and structural interconnectedness of areas experiencing similar MetS effects.

Strengths:

The effects of metabolic syndrome on the brain are still incompletely understood, and such multi-scale analyses are important for the field. Despite the study's sole 'correlation-based' nature, it yields valuable results, including several scales of brain parameters (cortical thickness, cellular, and network-based). The results are robust and benefit from two 'large-scale' datasets, resulting in highly powered statistics.

Weaknesses:

However, some concerns arise regarding certain interpretations and claims made by the authors. In particular, it is not entirely convincing that the authors' results are relevant for studying insulin resistance as a clinical measure of MetS. This is due to the use of non-fasting glycemia as a metric, which the authors claim represents insulin resistance. While non-fasting blood glucose is a potential, albeit poor, indicator of insulin resistance, claiming a direct correlation between insulin resistance and cortical thickness does not seem entirely convincing. By doing so, the authors suggest that insulin resistance might have a weak contribution to cortical thickness abnormalities, with a rather low 'loading' of glycemia compared to the other MetS metrics, although this cannot be conclusively determined from these results.

Reviewer #2 (Public Review):

Summary:

In this manuscript, Petersen et al. aimed for a comprehensive assessment of the relationship between cardiometabolic risk factors and cortical thickness. They found that a latent variable reflecting higher obesity, hypertension, LDL cholesterol, triglycerides, glucose, and lower HDL cholesterol was associated with lower cortical thickness in orbitofrontal, lateral prefrontal, insular, anterior cingulate, and temporal areas. In sensitivity analyses, they showed that this pattern replicated across cohorts and was also consistent with a clinical definition of the metabolic syndrome.

Further, when including cognition in the multivariate analysis, the pattern remained unchanged and indicated that cardiometabolic risk factors were associated with worse cognitive performance across different tests. The authors investigated the cell types implicated in the regions associated with cardiometabolic risk using the Allen brain atlas and found that the density of excitatory neurons type 8, endothelial cells, and microglia reliably co-located with the pattern of cortical thickness. Furthermore, they showed that cortical regions more strongly associated with MetS were more closely structurally & functionally connected than others.

Strengths:

This study performed a comprehensive assessment of the combined association of cardiometabolic risk factors and brain structure and investigated micro- and macroscopic underpinnings. A major strength of the study is the methodological approach of Partial Least Squares which allows the authors to not single out risk factors but to take them into account simultaneously. The large sample size from two cohorts allowed for different sensitivity analyses and convincing evidence for the stability of the first latent variable. The authors demonstrated that the component was also reliably related to cognitive performance,

replicating multiple previous studies that evidenced associations of different components of the MetS with worse cognitive performance.

The novel contribution of the study lies in the virtual histology and brain topology investigation of the cortical pattern related to MetS. The virtual histology provided clear evidence of the co-localization of endothelial, glial, and excitatory neuronal cells with the regions of MetS-associated cortical thinning while the brain topology analysis highlighted the disproportionate structural and functional connectivity between associated regions. This analysis provides insights into the role of inflammatory processes and the intricate link between gray matter morphology and microvasculature, both locally and in relation to long-range connectivity. This information is valuable to inform future mechanistic studies.

Weaknesses:

The study is exclusively cross-sectional which does not allow to the authors to disentangle causes from consequences. While studies indicate that most of the differences seen in middle age are probably consequences of the MetS on the vasculature, blood-brain barrier, or inflammatory processes, differences in cortical morphology might also represent a risk factor for weight gain.

Another limitation is the omission of subcortical structures and the cerebellum which might have provided additional information on the pattern of GM differences associated with MetS.

The study is exploratory in nature and for the contextualization analyses it is difficult to judge whether those were selected from a larger pool of analyses. The analysis approach taken to relate the cardiometabolic risk, brain structure, and cognition does not allow the reader to determine whether brain regions most strongly related to the MetS are the ones also most strongly associated with cognitive performance. The cortical pattern arising from the models including cognition is not thoroughly compared to the MetS-only pattern and therefore, it is difficult to estimate to which extent the MetS-related cortical patterns explain variance in cognitive performance.

Reviewer #3 (Public Review):

Summary:

This study investigates the continuous effect of MetS components - namely, obesity, arterial hypertension, dyslipidemia, and insulin resistance - on cortical thickness. It also examines the spatial correlations between MetS effects on cortical thickness with brain cellular and network topological attributes. Additionally, the authors attempt to explore the complex interplay among MetS, cognitive function, and cortical thickness.

The results reveal a latent relationship between MetS and cortical thickness based on a clinical-anatomical dimension. Furthermore, the effect of MetS on cortical thickness is linked to local cell types and network topological attributes. These findings suggest that the authors achieved most, though not all, of their research objectives.

The conclusions are mostly well supported by data and results. However, the use of "was governed by" in the conclusion section suggests a causal relationship. This phrasing is inappropriate given that the study primarily employs correlational analyses.

Strengths

The study presents several strengths:

This study undertakes a comprehensive assessment encompassing the full range of MetS components, such as obesity or arterial hypertension, rather than adopting a case-control study approach (categorizing participants into MetS or non-MetS groups) as seen in some previous research. Utilizing Partial Least Squares (PLS) for correlational analysis effectively

addresses issues of multicollinearity (or high covariance among MetS components) and explores the relationship between MetS and brain morphology.

The study leverages two datasets, examining a large sample size of 40,087 individuals. This substantial sample potentially aids in identifying nuanced and underexplored brain anomalies. By incorporating high-quality MRI images, standardized data, and statistical analysis procedures, as well as sensitivity analyses, the results gain robustness, which addresses the limitations of small samples and low reproducibility.

In the context of MetS, this research uniquely employs the concept of imaging transcriptomics, i.e. virtual histology analysis. This approach allows the study to explore intricate relationships between cellular types and cortical thickness anomalies.

Weaknesses

While this work has foundational strengths, the analyses and data seem inadequate to fully support the key claim and analysis. In particular:

After a thorough review of the methods and results sections, I found no direct or strong evidence supporting the authors' claim that the identified latent variables were related to more severe MetS to worse cognitive performance. While a sub-group comparison was conducted, it did not adequately account for confounding factors such as educational level. Additionally, the strength of evidence from such a sub-group comparison is substantially weaker than that from randomized controlled trials or longitudinal cohort studies. Therefore, it is inaccurate for the authors to assert a direct relationship between MetS and cognitive function based on the presented data. A more appropriate research design or data analysis approach, such as mediation analysis, can be employed to address this issue.

The use of the imaging transcriptomics pipeline (virtual histology analysis) to explore the microscale associations with MetS effects on the brain is commendable and has shown promising results. Nevertheless, variations in gene sets may introduce a degree of heterogeneity in the results (Seidlitz, et al., 2020; Martins et al., 2021). Consequently, further validation or exploratory analyses utilizing different gene sets can yield more compelling results and conclusions.

Author Response

We highly appreciate the constructive feedback provided by the reviewers, which we believe will greatly improve the quality of our work. We were encouraged to see that our manuscript was considered to be “important”, of “great interest” as well as to “yield valuable results”.

We also highly appreciate the overall positive eLife assessment. However, we were surprised to read that our “results range from solid from inadequate”. This especially applies given the positive and engaging nature of the reviews which seem to mainly concern the results interpretation being “inadequate” rather than the results themselves. Hence, we kindly request a reconsideration of this aspect of the assessment.

Moreover, there is one Reviewer comment we would like to address directly. Reviewer #3 pointed out that “this study did not conduct a direct association analysis between MetS and cognitive levels without considering subgroup comparisons.” and that “After a thorough review of the methods and results sections” she/he “found no direct or strong evidence supporting the authors' claim that the identified latent variables were related to more severe MetS to worse cognitive performance. While a sub-group comparison was conducted, it did not adequately account for confounding factors such as educational level.”.

We appreciate the observations of Reviewer #3 regarding the absence of a direct association analysis between Metabolic Syndrome (MetS) and cognitive levels without subgroup comparisons, and the lack of evidence linking latent variables to MetS severity and cognitive

performance. Our apologies for any confusion caused by unclear presentation. Our study incorporated association analyses between MetS, brain structure, and cognition using MetS components, regional cortical thickness, and cognitive performance data in a PLS. These analyses were separately performed on the UK Biobank and HCHS datasets, due to their distinct cognitive assessments. We adjusted for age, sex, and education in the subgroup analyses by removing their effects from the input variables. The primary latent variables demonstrated significant associations with MetS components, cortical thickness, and cognitive scores, indicating that higher obesity, blood pressure, lipidemia, and glycemia levels correlate with lower cognitive performance. These relationships are detailed in supplementary figures S15b and S16b, with negligible loadings for age, sex, and education, confirming effective deconfounding. We acknowledge the reviewer's constructive feedback and will enhance the clarity of the Methods and Results sections, including conducting a mediation analysis.

Furthermore, we strive to incorporate the Reviewers' other suggestions into the analysis. The revision will include major changes to the manuscript.

In response to Reviewer #1:

- We will revise considering non-fasting plasma glucose as a surrogate marker of insuline resistance.
- We will report Field IDs of the used UK Biobank variables.
- We aim to moderate causal interpretations and reword the indicated passages for clarity.

In response to Reviewer #2:

- We will reconsider claims of binarizing vascular dementia and Alzheimer's dementia pathophysiology.
- We will further explore the cell type associations of the other latent variables.
- We will expand the discussion regarding conclusions from our results and the future outlook.

In response to Reviewer #3.

- We will add an additional flowchart to detail the virtual histology analysis.
- We will add a discussion of the second latent variable.
- We will conduct a mediation analysis to statistically assess the mediation effect of brain structure on the relationship between MetS and cognitive performance.

We are convinced that with these revisions, our manuscript will align even more closely with the high standards of eLife and make a strong contribution to its distinguished portfolio. We thank you for your consideration.